

University of Moratuwa

Department of Electronic and Telecommunication Engineering



EN2091– Laboratory Practice and Projects

Metal Detector

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Submitted in partial fulfillment of the requirements for the module
EN 2091 Laboratory Practice and Projects

Date: December 14, 2025

Abstract

The metal detector is an analog sensing and detection system that identifies metallic objects by monitoring perturbations in an electromagnetic field produced by a driven search coil. A square-wave oscillator implemented with the MCP6002 dual op-amp drives the search coil at a center frequency of 26 kHz (observed variation approximately 25–27 kHz). Metallic targets near the coil alter its effective inductance and/or loss, producing measurable changes in the oscillator waveform (amplitude and phase). An analog signal-conditioning chain consisting of rectification/envelope detection and low-pass filtering extracts the slowly varying amplitude signature caused by the target. A comparator stage—also implemented with the MCP6002—compares the filtered envelope against an adjustable threshold to generate a digital detect signal. Visual and audible outputs (LEDs, buzzer) provide user feedback. The design leverages the MCP6002’s rail-to-rail I/O and dual-channel convenience to integrate oscillator and comparator functions in a compact, low-power analog system. Key concerns addressed in the design include oscillator stability at 26 kHz, coil geometry and sensitivity, noise immunity, hysteresis in thresholding to prevent chatter, and PCB layout practices to minimize EMI and cross-coupling. The project illustrates applied analog circuit design concepts such as inductive sensing, oscillator design, active filtering, comparator hysteresis, and system integration for real-world object detection.

Abbreviations and Acronyms

PCB	Printed Circuit Board
OpAmp	Operational Amplifier
LED	Light Emitting Diode
BFO	Beat Frequency Oscillator
GBW	Gain-Bandwidth Product

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1 Introduction and Functionality

A current carrying inductor coil creates magnetic flux lines. When a metal is brought closer to such a field eddy currents form in the said metal to oppose the movement of it. This is the basic principle used in any metal detector.

Our **Metal Detector** uses the **BFO (Beats Frequency Oscillator)** principle. A typical BFO setup consists of two oscillators: 1. The **search oscillator** 2. The **reference oscillator**. The frequencies of the two oscillators are adjusted so that their values are nearly the same. The two frequencies are mixed to generate the beat frequency (the difference) which drives the speaker after some amplification.

The search oscillator contains an inductor coil. When a metal object is brought closer to it, its inductance value changes, thus changing its frequency. The difference frequency changes because of this, with which we identify the presence of a metallic object.

2 System Architecture

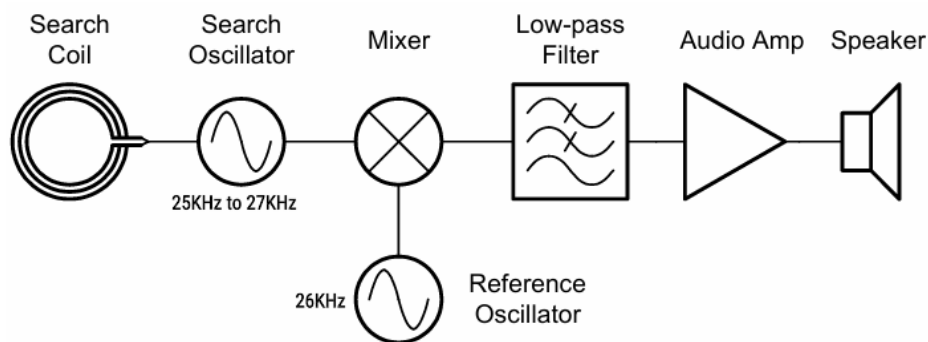


Figure 1: Functional Block Diagram

2.1 Search Coil

A major design decision for our product was regarding the operating frequency of the search oscillator.

We needed our product to be able to distinguish between ferrous and non-ferrous metals. The reason for the difference between the two lies in their atomic structure. However, frequency of the search oscillator affects this ability of the BFO detector. When current with a high frequency is flowing in a conductor it has a tendency to move along the outside surface. This phenomenon is known as the "skin effect". The higher the frequency the more pronounced this effect becomes. If the magnetic field generated by the search coil only penetrates the metal target to a shallow depth, then the true nature of the target may be misinterpreted and ferrous objects can be mistaken for non-ferrous, and vice-versa. This effect starts to be noticeable above 100kHz. Therefore, we wanted the operating frequency to be below 100kHz.

Other concern was the sensitivity. Consider the following equation for resonance frequency of an oscillator circuit formed using an inductor and a capacitor.

$$f = \frac{1}{2\pi\sqrt{LC}} \quad (1)$$

Assuming when a metal is brought closer, the inductance value changes by 5%. Then we observe a frequency deviation of 0.024%. So, to obtain a greater beats frequency for a small change in inductance (i.e. higher sensitivity), higher operating frequency is ideal.

By replacing the speaker with a buzzer we were able to decide on **25kHz** for the frequency, taking the simplicity of the circuit into consideration as well.

In our team's setup the search coil is wound using 0.3 gauge copper wire. The coil was 10mm in length, 35mm in diameter and had 52 wounds in a circular plastic holder. The inductance was measured to be around 800 μ H

2.2 Search Oscillator Circuit

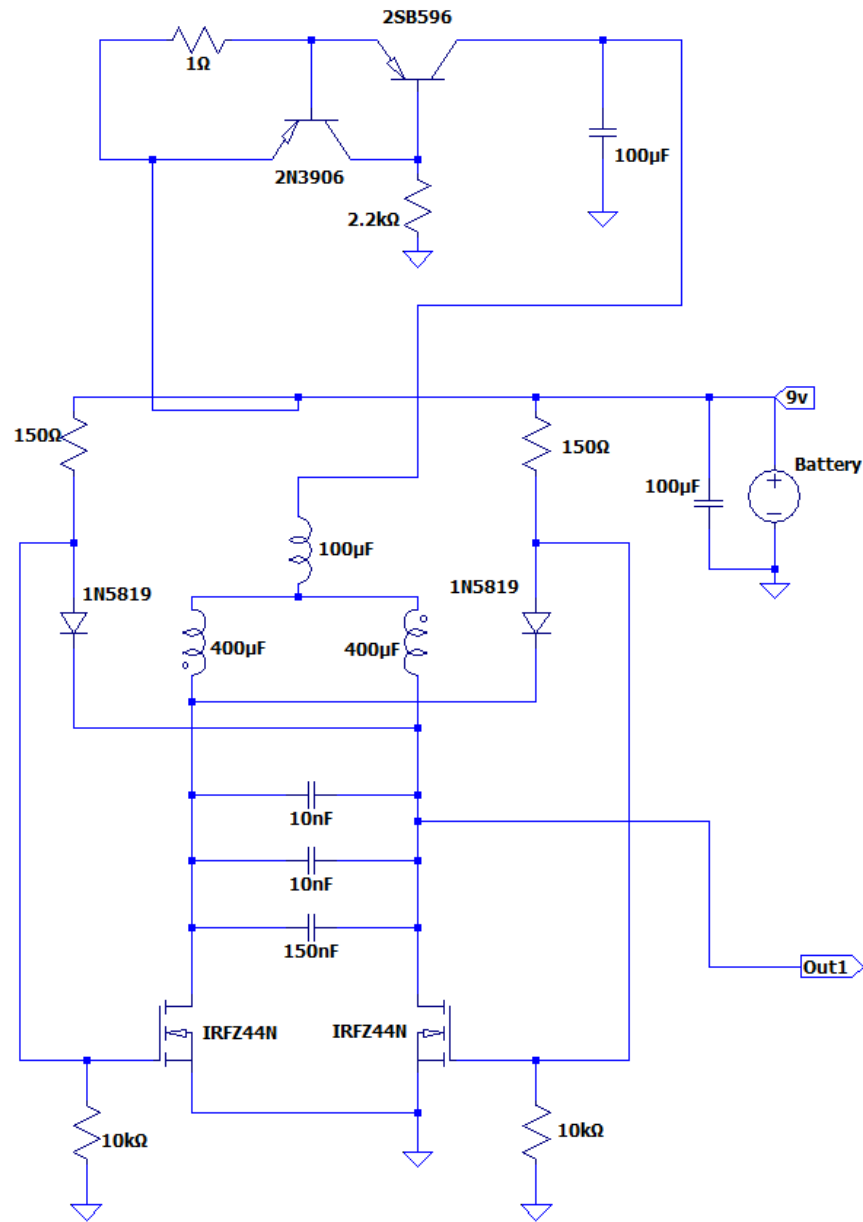


Figure 2: Search Oscillator Circuit

This is the circuit that holds the search oscillator. On top, a current limiter circuit is used to ensure safety of the circuit. The inductor coil is set up in a center tapped configuration to get a half sinusoidal output from this apparatus. This ensures the circuit is symmetric and become less prone for instabilities.

2.3 Reference Oscillator Circuit

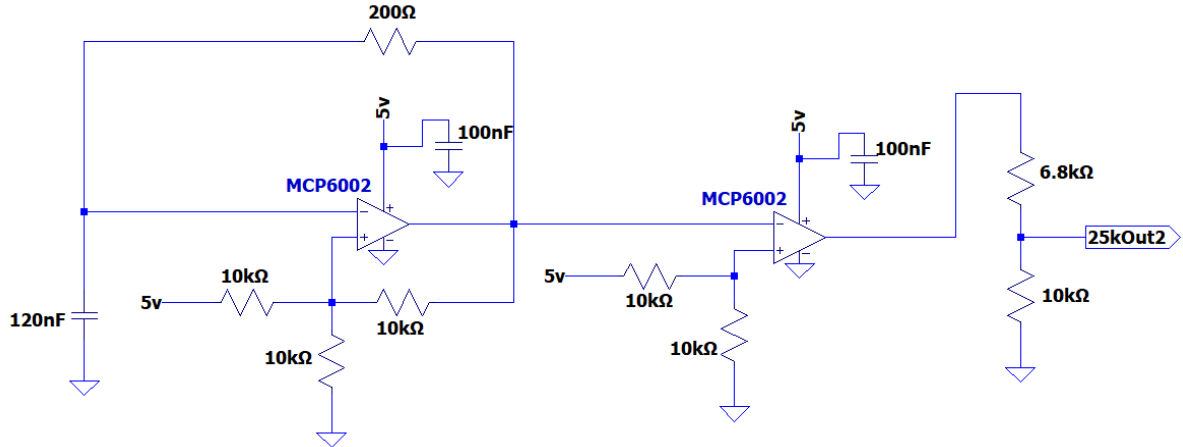


Figure 3: Reference Oscillator Circuit

The reference oscillator is a standard hysteresis square waveform generator implemented using OpAmps. The output is a **5v, 25kHz** near square wave.

For a better shape of the square waveform another OpAmp was placed serially in comparator mode. **MCP6002** was used for its **ability to work with single power supply, rail-to-rail output** and **moderate slew rate**, which are crucial factors in producing a better square wave.

For tuning the circuit two potentiometers were used in parallel with 100Ω resistor between the feedback loop of the OpAmp. 5kΩ and 10kΩ for fine and coarse tuning, respectively.

2.4 Mixer Circuit

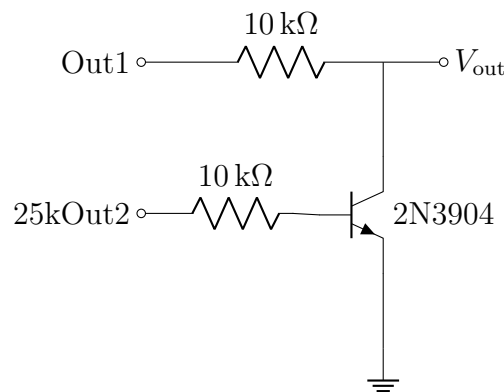


Figure 4: Mixer Circuit

Out1 is a half sinusoidal waveform around 25kHz output from the center tapped inductor coil in the search oscillator. **25kOut2** is the output of the reference oscillator. Due to the square pulse output, the transistor acts as a switch which therefore acts as a multiplying circuit.

Due to infinite harmonics in a square waveform, the resulting output waveform of the mixer circuit contains infinite frequencies.

2.5 Low-pass Filter Circuit

We filter out only the frequencies of the audible range using a **first order low pass filter**. A first order filter is enough for the design, as the next closest frequency is around 50 kHz (the sum of the two frequencies of the mixer).

$$f_c = \frac{1}{2\pi RC}$$

$$f_c = \frac{1}{2\pi(22 \times 10^3)(10 \times 10^{-9})}$$

$$f_c \approx 723 \text{ Hz}$$

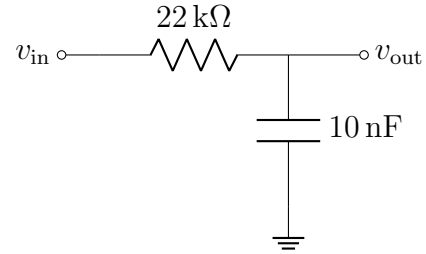


Figure 5: Low-pass Filter Circuit and Cutoff Frequency Calculation

2.6 Fixed Length Square Pulse Shaper

The output from the low pass filter can be directly amplified to drive a speaker. However, this idea has some pitfalls:

1. **Out of Audible Range:** Due to relatively low operating frequency, beats frequencies may not lie in the audible range for certain detections. Making it difficult for users to detect presence of a metal.
2. **Variable Amplitude:** The amplitude of the beat frequency signal depends on the size and proximity of the target. Small or distant objects produce very weak signals that may remain inaudible even after linear amplification.
3. **Lack of Defined Threshold:** Without a shaping stage, there is no distinct “trigger sound.” This creates ambiguity, making it difficult for the user to distinguish between a genuine detection and random signal drift.

Addressing this issues, we decided on driving a buzzer using fixed length pulses. With this design, higher changes of the frequency will be projected as pulses with low time period. Therefore higher the clicks of the buzzer implies a high intensity metal object has been detected.

The following circuit achieves this by sending the low pass filter output through a hysteresis comparator to filter out noise and create a square waveform which is then sent through a high pass filter. The output is a **train of decaying exponentials**. When this waveform is sent through another hysteresis comparator, shapes the pulse we need to drive the buzzer. See [Figure 24]. The first order high pass is used to determine the length of the pulse. Higher the cutoff frequency, lesser the pulse length.

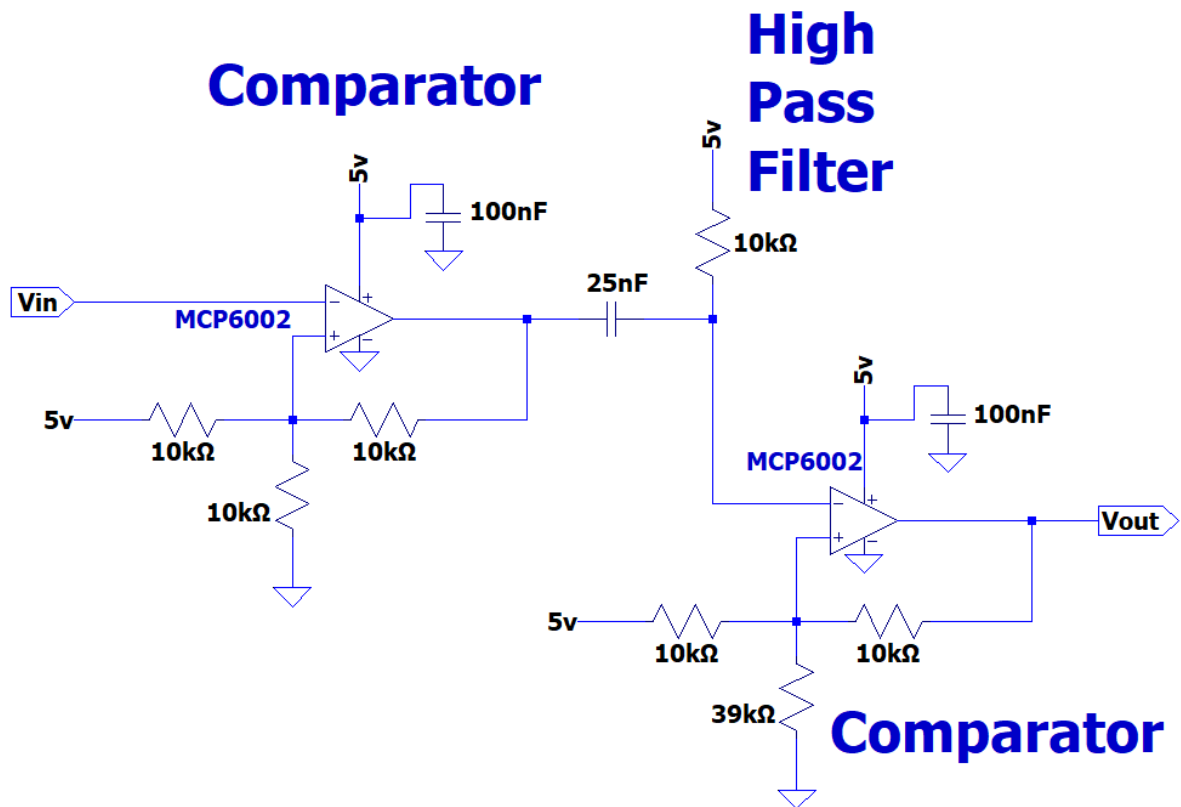


Figure 6: Fixed Length Square Pulse Shaper

2.7 Buzzer Amplifier Circuit

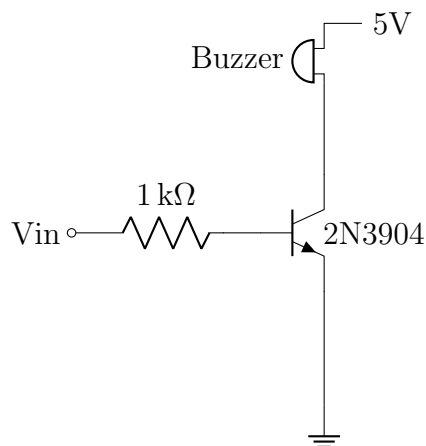


Figure 7: Buzzer Amplifier Circuit

2.8 9v to 5v Regulator

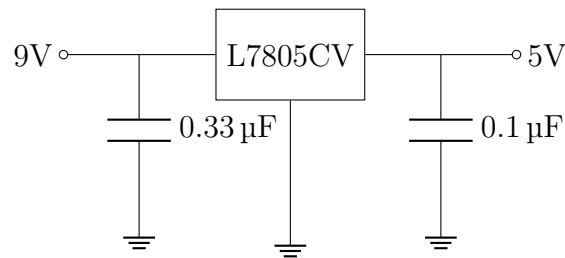


Figure 8: 9V to 5V Conversion

3 Component Selection (Main electronic components)

3.1 MCP6002 OpAmp

- **Dual Op-Amp:** Two amplifiers per IC for oscillator and comparator use
- **Rail-to-Rail I/O:** Full swing on single-supply designs (0-5V)
- **Wide Supply Range:** Operates from 1.8V to 6V for battery-powered systems
- **Single Power Supply operation:** Operates with a single power supply, made the circuit design much simpler
- **Low Power:** Only 100 μA typical quiescent current (ideal for portable detectors)
- **Slew Rate:** 0.6 V/ μs (edges moderately rounded at 25 kHz)
- **Bandwidth:** 1 MHz GBW suitable for low-frequency analog stages

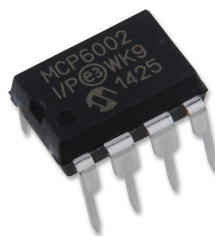


Figure 9: MCP6002 Operational Amplifier

4 PCB Design

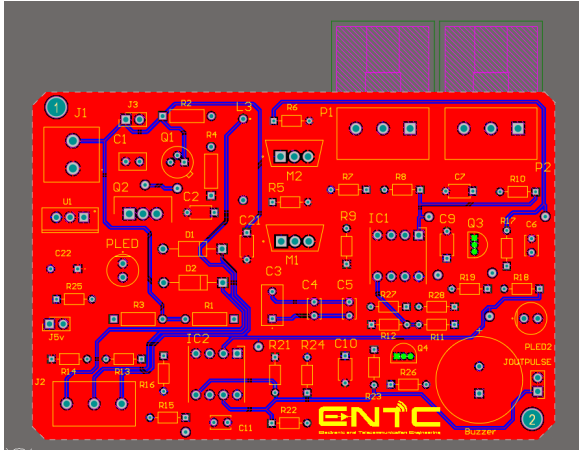


Figure 10: Top Layer View

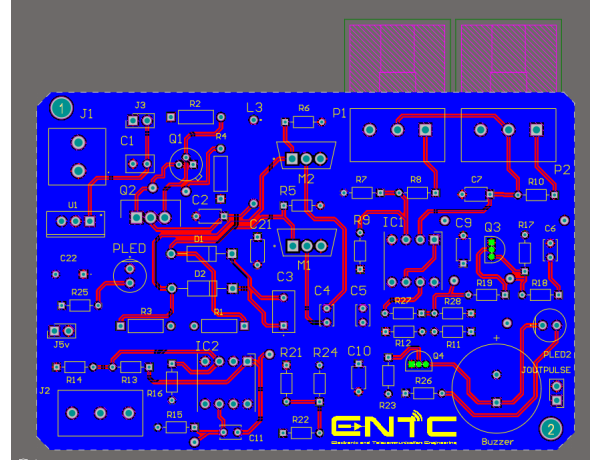


Figure 11: Bottom layer view

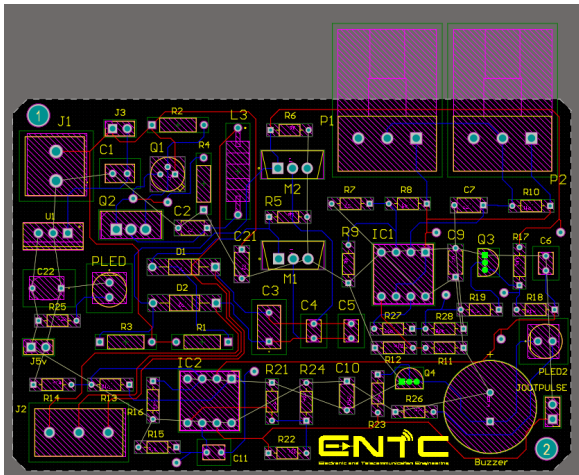


Figure 12: 2D pathway of the Circuit

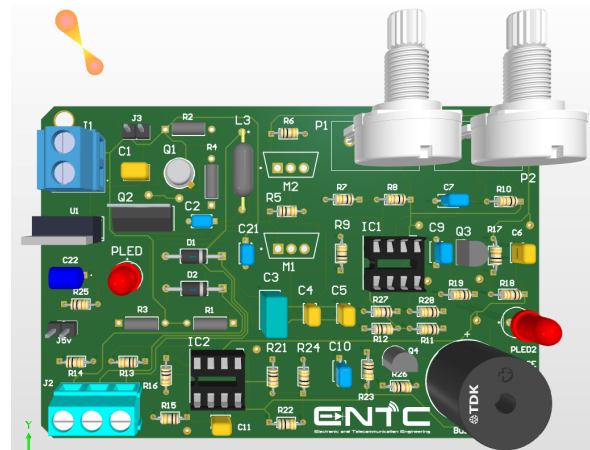


Figure 13: 3D view of the Circuit



Figure 14: Soldered PCB

5 Enclosure Design

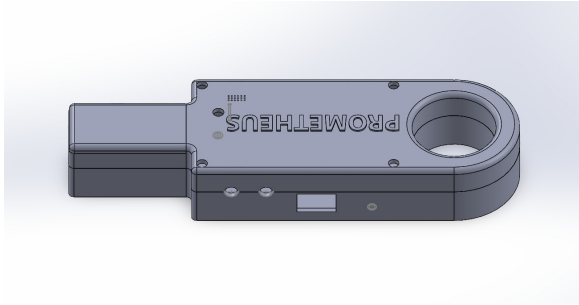


Figure 15: Front View

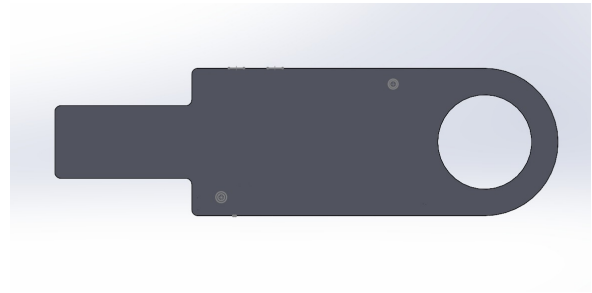


Figure 16: Back view

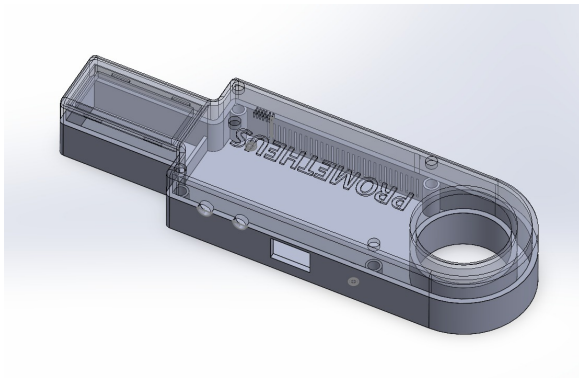


Figure 17: Right side view

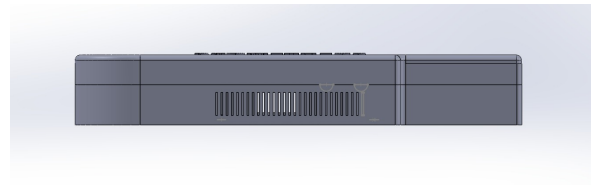


Figure 18: Left side view



Figure 19: Final Product

6 Software Simulation and Hardware Testing

6.1 Software Simulation (Using LTspice)

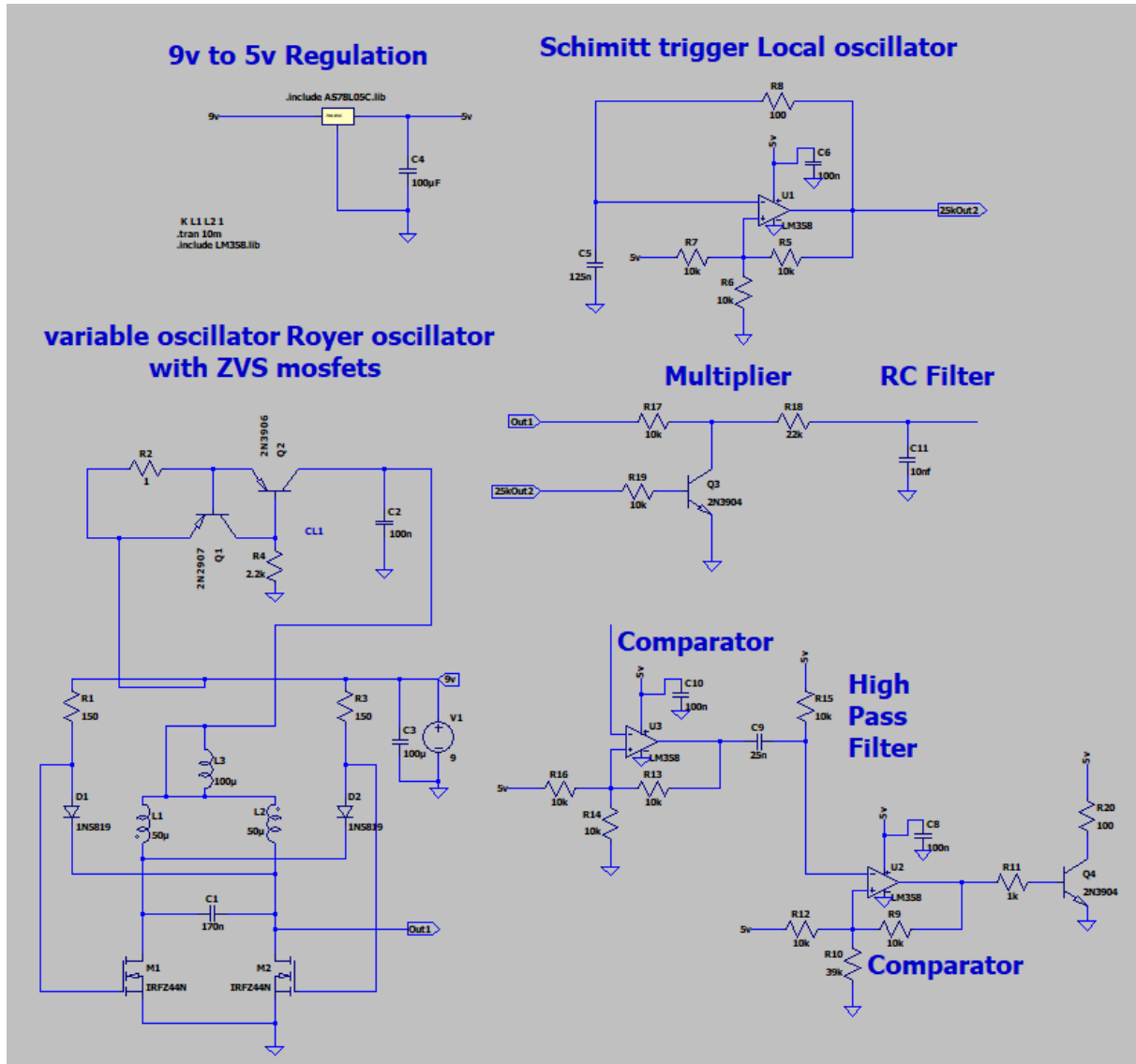


Figure 20: Circuit Diagram

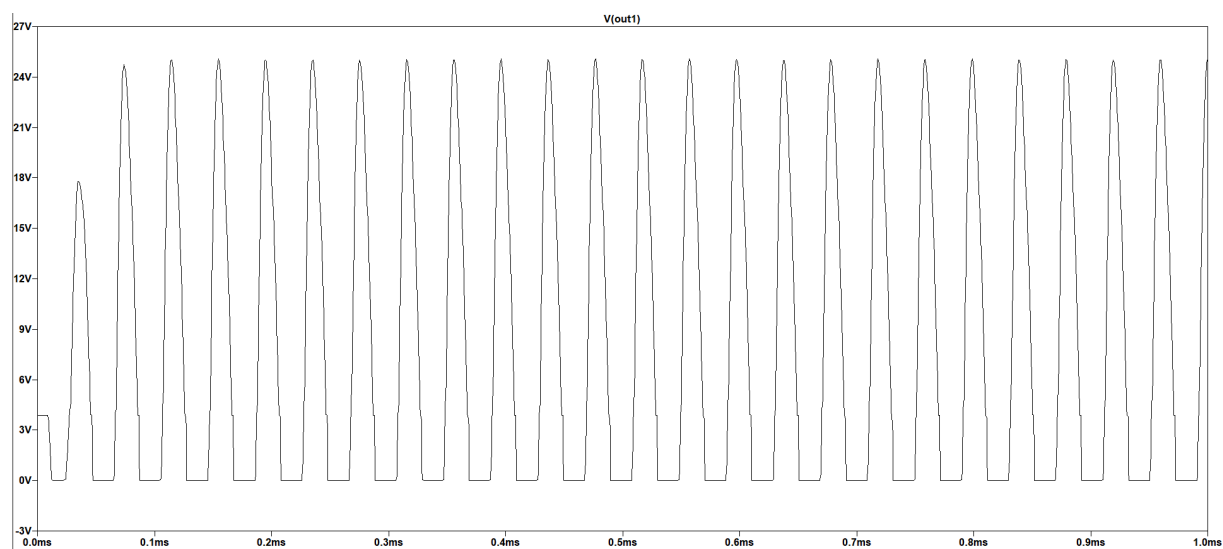


Figure 21: Out1 Waveform (Around 25KHz)

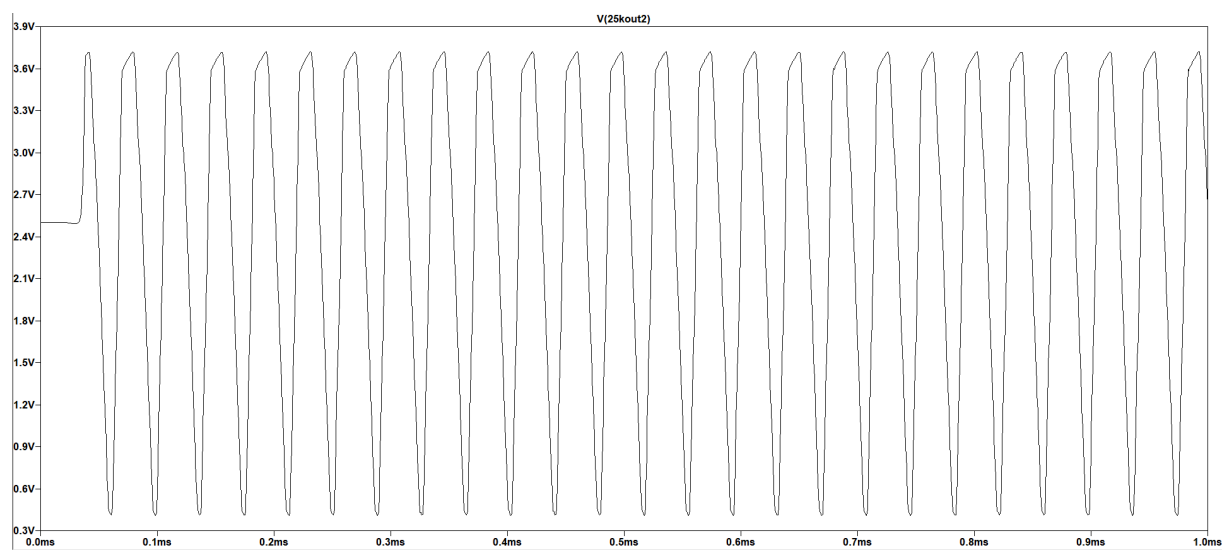


Figure 22: 25kOut2 Waveform

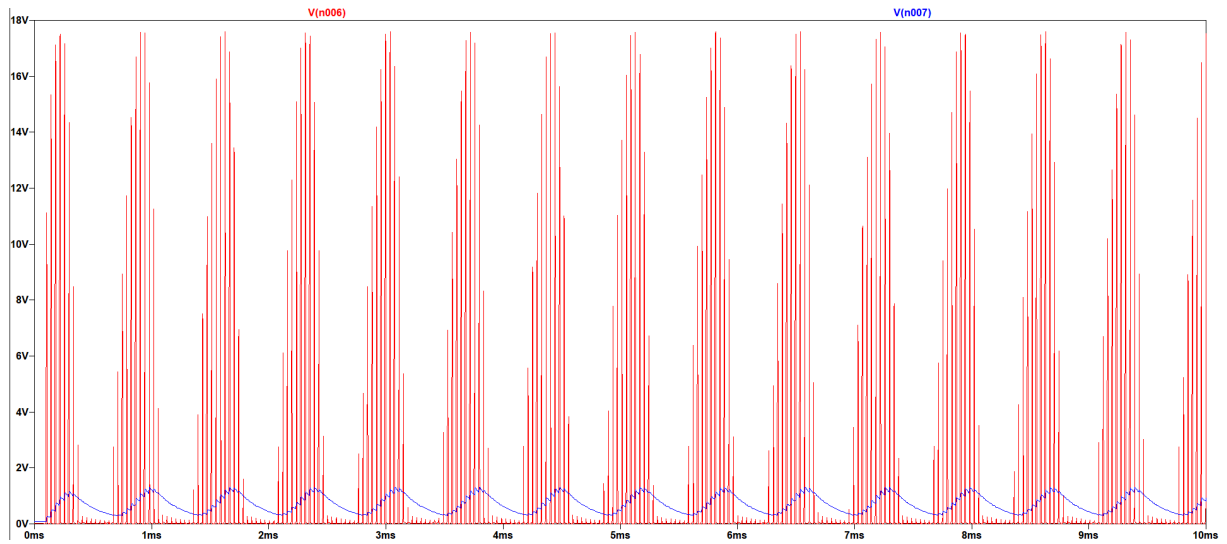


Figure 23: Multiplier output (ORANGE) and Low pass filter output (BLUE)

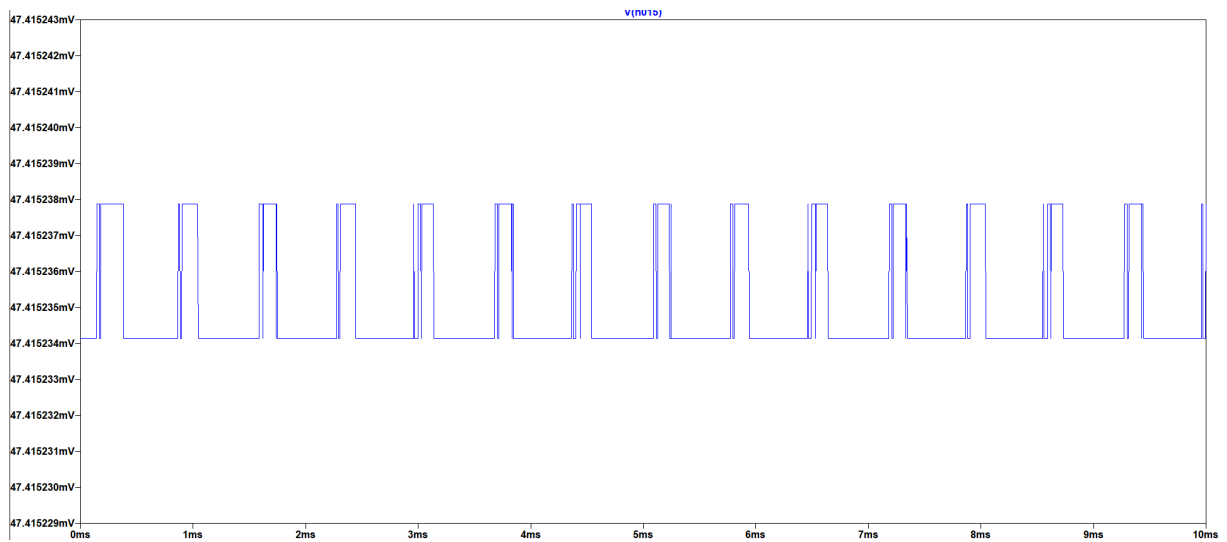


Figure 24: Output Sound Pulse (1KHz)

6.2 Hardware Testing

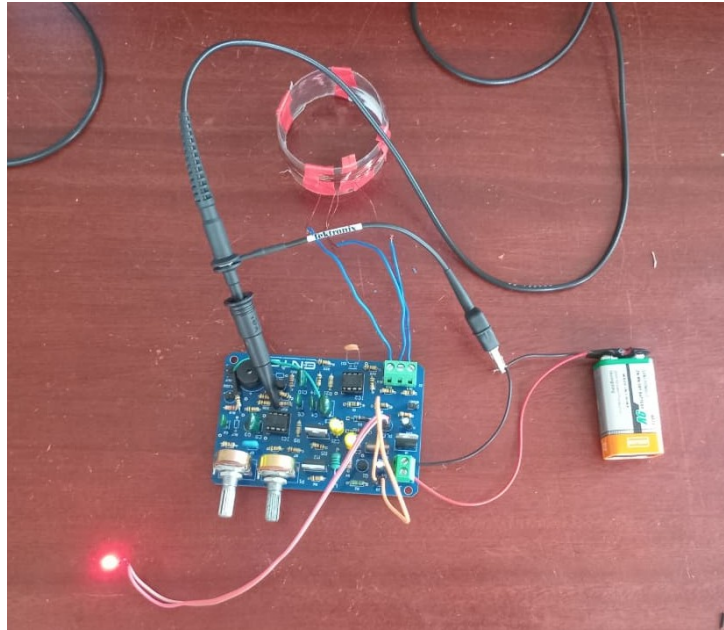


Figure 25: PCB Testing

7 Conclusion & Future Works

A fully functioning **Handheld Portable Analog Metal Detector** was realized at the end of EN2091 Analog Project. The complete product is able to successfully detect a metal of the size of a coin within a maximum distance of 30mm. Furthermore, it is able to distinguish between ferrous and non-ferrous metals.

There are multiple aspects that can be further improved on this design. Firstly, the drifting of the reference oscillator due to accumulated heat causes the tuning to be slightly off with time. This can be improved by employing crystal oscillators to generate the square wave. Furthermore the detection range capabilities can be further increased by: increasing the operating frequency and employing a inductor coil with larger surface area.

8 Contribution of Group Members

Student's Name (Index No.)	Contribution
Samarasinghe S.M.R.R. (230566U)	Voltage Comparator, High-Pass Filter Designing, Component Selection and Testing
Eranga W.A.O. (230175U)	Electromagnetic Inductor, Search Coil Oscillator Designing, PCB Designing, and Soldering
Tharushika G.K.E. (230636K)	Voltage Multiplier and RC Filter Designing, PCB Designing, and Enclosure Designing
Gamage S.K. (230195F)	Schmitt Trigger Oscillator Designing, Enclosure Designing, System Testing, and Optimization

Acknowledgment

We would like to sincerely thank **Mr. Kajanan**, our project supervisor, for his unwavering support and direction during the development of the Metal Detector project. His insightful recommendations during the initial phases of circuit design, combined with his invaluable guidance in identifying and resolving issues during the PCB and enclosure design stages, were instrumental in bringing this project to a successful and satisfactory conclusion.

Additionally, we would like to thank our lecturer in Signals and Systems, **Dr. Pranjeevan Kulasingham**, for giving us a thorough understanding of analog filter design concepts, which served as the basis for our filter implementation in this project.

We sincerely thank **Dr. Sampath Perera**, the EN2091 Laboratory Practice and Projects module coordinator, for his general guidance, support, and provision of the tools required to successfully finish this project.

Lastly, we would like to thank all of the laboratory assistants in the Department of Electronic and Telecommunication Engineering for their prompt advice, technical support, and help with the hardware testing and implementation phases.

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Appendix

PCB Schematics

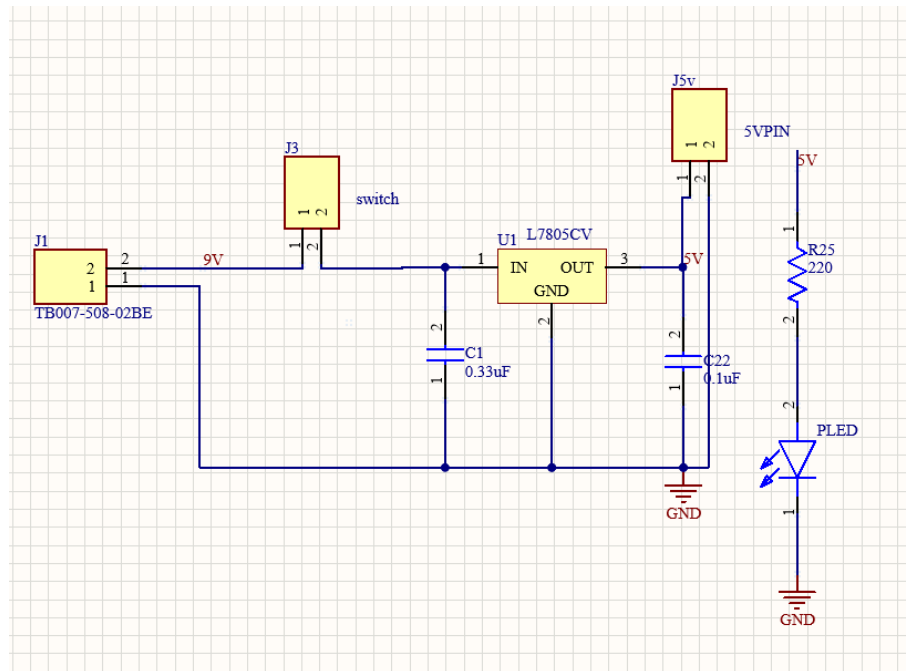


Figure 26: 9V to 5V Voltage Regulator

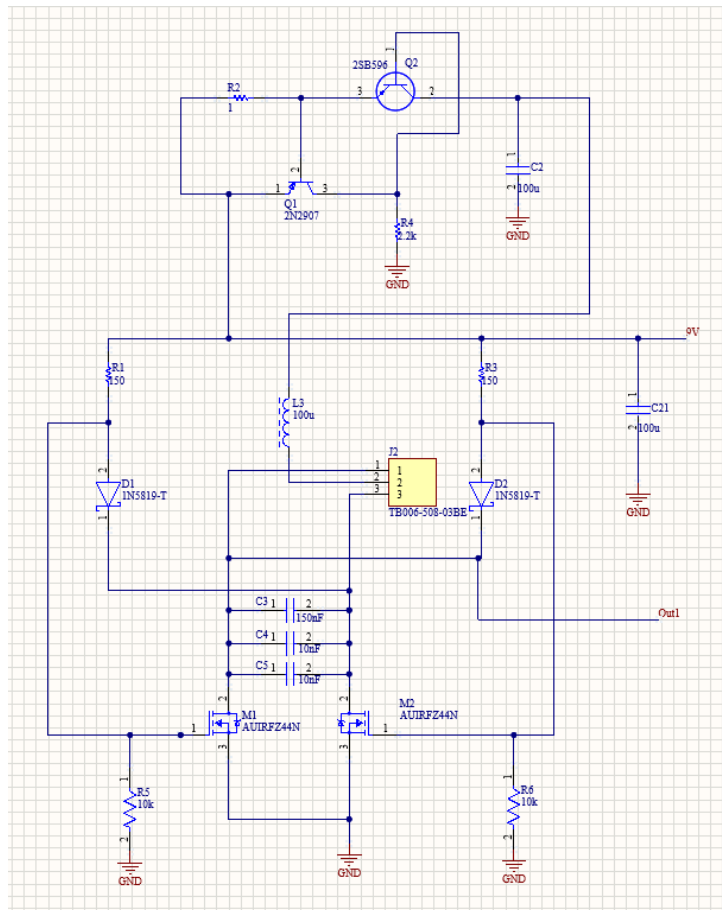


Figure 27: Search Oscillator

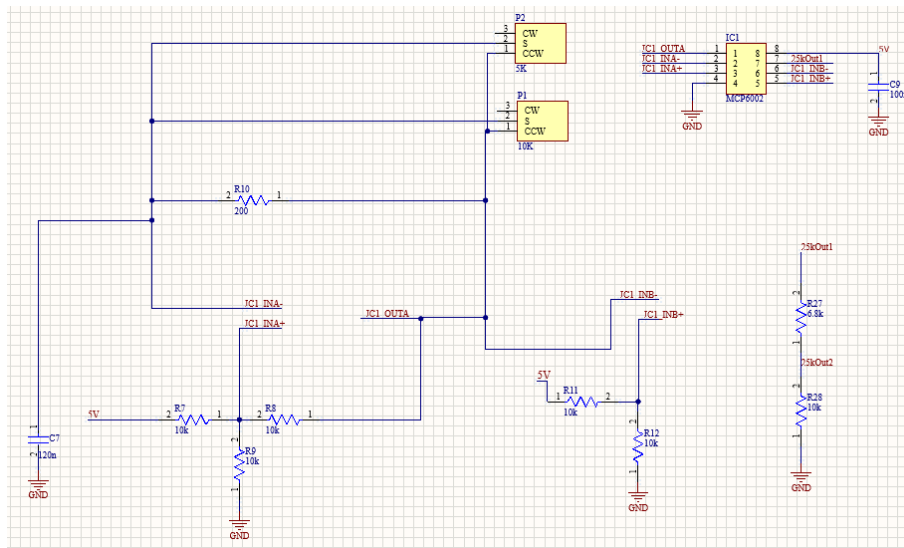


Figure 28: Reference Oscillator

